ModelIt!

Teacher's Guide

# How Do Plants Grow from Seeds?

The Science of the Plant Life Cycle

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 1 |
| **NGSS Standard** | 1-LS1-1 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: 1-LS1-1**

Use materials to design a solution to a human problem by mimicking how plants and animals use their external parts to help them survive, grow, and meet their needs.

## Three-Dimensional Learning

| | |
|---|---|
| Science Practice | **Planning and Carrying Out Investigations** — Students plant seeds and investigate how water affects plant growth over time. |
| Disciplinary Core Idea | **LS1.A Structure and Function** — Plants have different parts that help them survive, grow, and make more plants. |
| Crosscutting Concept | **Cause and Effect** — Students explore how providing or withholding water causes plants to grow or not grow. |

## CAST Testing Alignment

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every plant on Earth started as a tiny seed. Inside each seed is a baby plant waiting for the right conditions to start growing. Understanding how seeds grow helps us grow food and take care of our planet.

## What a Seed Needs

A seed is like a tiny survival kit. Inside it is a baby plant and a small amount of food to get it started. But the seed will not grow unless it gets three things from the outside: water, warmth, and eventually sunlight. When a seed absorbs water, it swells up and the hard outer coat softens. The baby plant inside begins to grow, sending a root down into the soil and a tiny stem up toward the light. This process is called germination.

## Stages of Plant Growth

After the seed germinates, the plant goes through several stages. First, a small sprout appears above the soil — this is called a seedling. The seedling grows a stem and its first leaves, which capture sunlight to make food through a process called photosynthesis. As the plant gets more sunlight, water, and nutrients from the soil, it grows taller and stronger. Eventually, many plants produce flowers, which can make seeds of their own. Then the cycle starts all over again!

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** — **Locate the System**
Students identify water amount as the external component they control (how much they water the plant) and plant growth as the internal result that changes.

**E** — **Establish Relationships**
Students discover that when water amount goes up (to a point), plant growth goes up — a positive relationship.

**V** — **Visualize & Model**
Students build a simple model showing how water amount connects to plant growth.

**E** — **Evaluate Outcomes**
Students test their model by comparing a watered seed to an unwatered seed and observing the difference.

**R** — **Revise & Extend**
Students reflect on how their model explains why farmers water their crops and gardeners tend their plants.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes. |
| **As a "what-if" tool** | When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Look at this tiny green sprout! It used to be just a little seed. How did it change? How does a seed know how to grow?"*

**Watch For**

Pass around some dry seeds. Ask: What do you think is inside this tiny thing? Could a whole plant really come from this?

## Slide 2: What Will We Learn?

3 min

**Say:** *"Today we will learn how a tiny seed becomes a big plant. We will learn the word SPROUT — that is when the baby plant first pokes out of the seed."*

**Watch For**

Show real sprouted seeds (or photos). Say: This is what a sprout looks like! It started as a seed just like the ones you are holding.

## Slide 3: Let's Be Scientists!

3 min

**Say:** *"How does a tiny seed turn into a big plant? What does the seed need to start growing?"*

**Watch For**

Think-pair-share. Students list what they think seeds need. Write ideas on the board.

## Slide 4: What Are the Parts?

5 min

**Say:** *"Our model has two parts: Water Amount is how much water we give the seed. Plant Growth is how tall and healthy the plant becomes."*

**Watch For**

Show two cups of soil with seeds — add water to one, leave the other dry. Say: We will check these every day!

## Slide 5: How Are They Connected?

5 min

**Say:** *"When we give the seed water, what do you think will happen? Will it grow or stay the same?"*

**Watch For**

Students use sentence frame: When water amount goes up, plant growth goes up. Draw the positive arrow together.

## Slide 6: What Did We Find Out?

5 min

**Say:** *"We found out that seeds need water to grow! The water helps the baby plant inside the seed wake up and start growing into a sprout."*

**Watch For**

Show time-lapse video or photos of a seed growing. Compare watered vs. unwatered results.

## Slide 7: STEM Challenge

10 min

**Say:** *"Now plant your OWN seed garden! Give it water, put it near sunlight, and track how it grows each day."*

**Watch For**

Distribute cups, soil, seeds, water. Students plant seeds and create a growth tracking chart to fill in daily.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Water Amount
- **INTERNAL:** Plant Growth

## Relationships

- **Water Amount to Plant Growth:** POSITIVE (+) — When a plant gets the right amount of water, it grows taller and healthier. Water helps the seed germinate and gives the plant what it needs to grow strong stems and green leaves.

## Key Discoveries

- Seeds need water, sunlight, and soil to start growing
- A plant grows from seed to sprout to a full plant with leaves and sometimes flowers
- Water is very important for plant growth — without it, the plant will not survive
- Plants grow taller and stronger when they get the right amount of water

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Seeds are not alive" | Seeds are alive, just sleeping! Inside every seed is a tiny baby plant that is waiting for water and warmth to wake up and start growing. When conditions are right, the seed comes to life and begins to grow. If seeds were not alive, they could never become plants. |

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Plants eat soil to grow" | Plants do not eat soil. They get nutrients from the soil through their roots, but most of their food comes from SUNLIGHT. Plants use sunlight, water, and air to make their own food in their leaves. Soil gives them a place to stand and some minerals, but the sun is their main food source. |